Exp: 8

Write the Python Program to Implement DFS.

Aim:

To write a python program to implement DFS.

Algorithm:

Step 1: Initialize a set to keep track of visited nodes.

Step 2: Call the DFS-Visit function with the starting node and the set of visited nodes.

Step 3: In the DFS-Visit function, add the current node to the set of visited nodes.

Step 4: For each neighbour of the current node, if it has not been visited yet, recursively call the DFS-Visit function with the neighbour node and the set of visited nodes.

Step 5: Repeat steps 3-4 until all reachable nodes have been visited.

Program:

```
# Define the graph as an adjacency list
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}
# Define the DFS function
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()
    visited.add(start) # mark the node as visited
    for neighbor in graph[start]:
```

```
        if neighbor not in visited:
            dfs(graph, neighbor, visited) # recursively visit unvisited neighbors
    return visited
# Call the DFS function with starting node 'A'
print(dfs(graph, 'A'))
```

Output:

```
============================== RESTART: D:/test.
{'C', 'F', 'B', 'E', 'D', 'A'}
```

Result:

Thus, the Program was executed and output found to be correct.